

YONAS TSEGAYE MEGRA, Ph.D.

Materials Scientist | 2D Materials | Molecular Electronics | Nanoelectronics

Advanced Materials Research Center (AMRC), Centre of Excellence for Molecular Innovation and Advanced Materials,

Academy of Life and Natural Sciences, Xi'an Jiaotong-Liverpool University (XJTLU), Suzhou, China

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RESEARCH PROFILE

Materials scientist and nanoelectronics researcher specializing in molecularly engineered two-dimensional materials, nanoscale interfaces, and charge-transport phenomena. My research integrates 2D-material synthesis and transfer, nanofabrication, molecular electronics, quantum transport, semiconductor devices, advanced materials characterization, and interfacial mechanics to understand and engineer structure–property relationships at the nanoscale. My trajectory has progressed from adhesion, cohesion, transfer, and interfacial mechanics of 2D materials to 2D semiconductor devices, molecular junctions, quantum interference, and advanced sensing. My long-term goal is to establish an independent research program developing molecularly engineered 2D interfaces for quantum electronics, low-power nanoelectronics, and intelligent sensing.

511

Google Scholar citations

482

citations since 2021

11

h-index

12

i10-index

RESEARCH THEMES

Molecularly Engineered 2D Interfaces — Molecular/2D-material interfaces, interfacial charge transfer, molecular anchoring, electronic coupling, defect engineering, and interface-driven functionality.

Quantum & Molecular Electronics — Single-molecule junctions, molecular conductance, quantum interference, nanoscale charge transport, and mechanically/electrically controlled molecular electronic states.

2D Semiconductor Nanoelectronics — Graphene, MoS₂, WS₂, WSe₂, 2D heterostructures, semiconductor devices, transistors, photodetectors, sensors, and emerging nanoelectronic architectures.

Nanoscale Interface Mechanics — Adhesion, cohesion, fracture, transfer, delamination, self-healing, and interfacial interactions in large-area 2D materials.

AI-Enabled Materials & Devices — AI-assisted materials research, data-driven analysis, computational approaches, intelligent sensing, and integration of advanced materials with emerging device architectures.

ACADEMIC APPOINTMENTS

Postdoctoral Research Fellow | June 2025–Present

Xi'an Jiaotong-Liverpool University, Suzhou, China — Advanced Materials Research Center (AMRC), Center of Excellence for Molecular Innovation and Advanced Materials, Academy of Life and Natural Sciences

- Conduct research on 2D-material-based molecular electronics, nanoelectronic devices, and advanced sensing platforms.
- Develop graphene–molecule and MoS₂–molecule hybrid junctions for nanoscale charge-transport studies.
- Investigate molecular interfaces, quantum transport, and interface-controlled electronic properties.
- Contribute to AMRC research development, including laboratory planning, infrastructure development, and the AI-Chemist research environment.
- Teach undergraduate laboratory courses, supervise SURF projects, and mentor MSc/PhD students.

Postdoctoral Researcher | April 2024–May 2025

Gwangju Institute of Science and Technology (GIST), Gwangju, Republic of Korea — Department of Semiconductor Engineering, Quantum Materials Interfaces and Nano Devices Laboratory

- Conducted research on 2D semiconductor materials, quantum-material interfaces, and AI-integrated sensing devices.
- Developed semiconductor and nanomaterial-based devices for electronic, optoelectronic, and sensing applications.
- Contributed to interdisciplinary projects involving materials science, semiconductor engineering, and artificial intelligence.

Postdoctoral Researcher | December 2022–March 2024

Sungkyunkwan University (SKKU), Suwon, Republic of Korea — Department of Smart-Fab Technology, Nanomaterial and Nano Manufacturing Laboratory

- Conducted research on nanomaterial synthesis, fabrication, and integration for advanced electronic and sensing applications.
- Investigated 2D-material properties, interfaces, and device performance through experimental studies.

Postdoctoral Researcher | September–November 2022
Sungkyunkwan University (SKKU), Suwon, Republic of Korea — School of Mechanical Engineering, Nanomaterial and Nano Manufacturing Laboratory

Research Assistant / Doctoral Researcher | August 2017–August 2022
Sungkyunkwan University (SKKU), Suwon, Republic of Korea — Nanomaterial and Nano Manufacturing Laboratory

- Developed fracture-mechanics-based methods for investigating adhesion and interfacial interactions of large-area 2D materials.
- Studied graphene, graphene oxide, and related 2D materials using mechanical testing and advanced characterization.
- Developed experimental and computational approaches for 2D-material transfer, bonding, delamination, and interface engineering.

Lecturer | September 2015–July 2017
Addis Ababa University, Addis Ababa, Ethiopia — School of Mechanical Engineering

- Taught Engineering Mechanics: Dynamics, Technical Drawing, and Maintenance of Machinery.
- Developed course materials, delivered lectures, supervised practical learning, and assessed student performance.

RESEARCH LEADERSHIP & INDEPENDENCE

- First-author leadership on research establishing crystallographic dependence of graphene adhesion on copper.
- First/co-first-author contributions to molecular-electronics and quantum-transport research.
- Development of research directions involving graphene–molecule and MoS₂–molecule hybrid junctions.
- Principal Investigator/supervisor for the 2026 XJTLU SURF undergraduate research project; co-supervisor for the 2025 project.
- Development and submission of competitive external research-funding proposals.
- Cross-disciplinary research spanning materials science, chemistry, mechanical engineering, semiconductor engineering, molecular electronics, and computational materials science.
- Contributions to laboratory infrastructure and research-environment development at XJTLU.

EDUCATION

Ph.D. in Mechanical Engineering | August 2017–August 2022
Sungkyunkwan University, Republic of Korea — Dissertation: “Study on the Interfacial Interactions of Large-Area Two-Dimensional Materials Using Fracture Mechanics”

M.Sc. in Mechanical Engineering — Mechanical Design | September 2015–July 2017
Addis Ababa University, Ethiopia

B.Sc. in Mechanical Engineering — Mechanical Design | September 2010–July 2015
Addis Ababa University, Ethiopia

RESEARCH FUNDING & GRANT DEVELOPMENT

PROGRAM	REQUESTED	STATUS
National Natural Science Foundation of China (NSFC)	RMB 300,000 / 3 years	Submitted; under review
China Postdoctoral Science Foundation — Special Grant, 19th Batch	RMB 180,000 / 1 year	Submitted; under review
Jiangsu Excellent Postdoctoral Program	RMB 150,000 / 1 year	Submitted; provincial review

TEACHING & MENTORING

- CHE103 — Introductory Inorganic Chemistry Laboratory, XJTLU (2025/26–Present): approximately 55 second-year undergraduate students; ten-week program; approximately 40 teaching hours; approximately 165 laboratory reports marked.
- Guided students in experimental procedures, calculations, data analysis, scientific interpretation, and conclusions; contributed to laboratory-manual and assessment development.
- SCI004 — Horizons in Science: invited lectures “The Materials Universe: From Atoms to Applications” and “Materials for a Sustainable Future: Energy, Environment and Circular Economy.”
- SURF 2026 — Principal Investigator: supervised three undergraduate researchers on “Engineering MoS₂–Molecule–Au Single-Molecule Junctions for Nanoscale Charge Transport Investigation.”
- SURF 2025 — Co-Supervisor: co-supervised two undergraduate researchers on quantum interference in 1,8-/1,10-alkanedithiol molecular junctions.

- Provide MSc/PhD mentoring in experimental design, 2D-material synthesis and transfer, molecular-junction fabrication, characterization, electrical measurements, data analysis, and scientific writing.

TEACHING INTERESTS

Materials Science & Engineering • Nanotechnology • 2D Materials • Electronic Materials • Semiconductor Materials & Devices • Molecular Electronics • Quantum & Molecular Electronics • Surface & Interface Science • Materials Characterization • Mechanics of Materials • Energy Materials • Sustainable Materials

Potential new courses: Molecular and 2D Materials for Nanoelectronics; Interfaces and Surfaces in Advanced Materials; Quantum and Molecular Electronics; Nanomaterials Characterization and Devices; Materials for Sustainable Energy Technologies.

SELECTED JOURNAL PUBLICATIONS

2026

1. **Megra, Y. T.**, Lim, S., Suk, J. W. "Crystallographic dependence of adhesion energy of chemical vapor deposition-grown monolayer graphene on copper foils." *Carbon* 247 (2026), 120948.
2. Fan, Y.†, **Megra, Y. T.†**, Tekalgne, M. A., Li, J., Liu, C., Dappe, Y. J., Yang, L. "Electrochemical Modulation of Destructive Quantum Interference in Meta-Connected Oligo(phenylene Ethynylene) Molecular Junctions." *Journal of Physical Chemistry C* 130 (20) (2026), 7269–7277. [Equal contribution.](#)
3. Ren, J., Zhang, J., Zhang, J., Li, J., **Megra, Y. T.**, Wang, J., Du, Y., Huang, R., Cui, Y., Liu, C., Yang, L. "Enhancing Anode-Free Lithium Metal Batteries: A Lightweight MPTS-MXene Current Collector for Superior Performance and Energy Density." *Energy & Environmental Materials*, e70352 (2026).
4. Zhang, J., Li, J., Ren, J., Zhang, J., Zhang, S., Huang, H., Yi, R., Yang, S., **Megra, Y. T.**, Liu, C., Yang, L. "Confined Ion Regulation via Synergy of Porosity and Acidity in Molecular Sieves for Lithium-Metal Batteries." *Advanced Functional Materials* (2026).
5. Tekalgne, M. A., Liu, C., **Megra, Y. T.**, Yang, L. "Layered Double Hydroxide@Carbon Composites in Lithium-Sulfur Batteries: Synthesis Strategies, Shuttle Suppression Mechanisms, and Applications." *Battery Energy* 5 (2) (2026).
6. Li, J., Huang, H., Liu, C., Liu, C.*, Zhang, S., Chen, Y., **Megra, Y. T.**, Zeng, Z., Zhang, Q., Zhao, C., Dappe, Y. J., Nichols, R. J., Yang, L. "Electric Field-Driven Dynamic Junction in Transition Metal Complexes for High-Contrast Hysteresis." *smartMat*. Accepted; forthcoming.
7. **Megra, Y. T.**, Tekalgne, M. A., Li, J., Liu, C., Dappe, Y. J., Yang, L. "Graphene-Molecule Hybrid Junctions: From Quantum Transport to Functional Devices." *Nano Materials Science*. Accepted; forthcoming.
8. **Megra, Y. T.†**, Tekalgne, M. A.†, Mamlouk, M., Li, J., Liu, C., Dappe, Y. J., Yang, L. "Charge Transport Across Defect-Engineered MoS₂-Molecule Interfaces in Single-Molecule Junctions." Under review.
9. Lee, S.-H.†, **Megra, Y. T.†**, Min, J.†, Jeong, M.-H., Woo, H. J., Han, J., Ha, J., Jang, G., Song, Y. J., Ra, H.-S.*, Yoon, H. H.*, Lee, J.-S.* "Multidimensional-Encoded High-Fidelity Spectrometers with a Floating-Base Bipolar Junction." Under review. [Equal contribution.](#)

2025

10. Yoon, H. H.†, Park, J. Y.†, **Megra, Y. T.†**, Baek, J. H.†, Song, M., Akinwande, D., Ha, D.*, Kang, D.-H.*, Shin, H.-J.* "Enabling the Angstrom Era: 2D Material-Based Multi-Bridge-Channel Complementary Field Effect Transistors." *npj 2D Materials and Applications* 9 (2025), 68. [Equal contribution.](#)

2024

11. Seok, H., Kim, M., Cho, J., Son, S., **Megra, Y. T.**, Lee, J., Nam, M. G., Kim, K.-W., Aydin, K., Yoo, S. S., Lee, H., Kanade, V. K., Kim, M., Mun, J., Kim, J. K., Suk, J. W., Kim, H.-U.*, Yoo, P. J.*, Kim, T.* "Electron Release via Internal Polarization Fields for Optimal S-H Bonding States." *Advanced Materials* (2024), 2411211.
12. Yamamoto, G., Oshima, K., Ramadhan, R. A., Lim, T., **Megra, Y. T.**, Suk, J. W., Watanabe, J., Okuda, H., Tanaka, F. "Tensile Strength Prediction of Unidirectional Polyacrylonitrile (PAN)-Based Carbon Fiber Reinforced Plastic Composites Considering Stress Distribution Around Fiber Break Points." *Composites Part A: Applied Science and Manufacturing* 183 (2024), 108234.
13. Lee, H.-J., Na, S. C., Lim, T., Yun, J., **Megra, Y. T.**, Oh, J.-H., Jeong, W., Lim, D., Suk, J. W.* "Vertical Graphene-Decorated Carbon Nanofibers Establishing Robust Conductive Networks for Fiber-Based Stretchable Strain Sensors." *Journal of Materials Science and Technology* (2024).

2023

14. Yoo, H. S., Kim, S. J., **Megra, Y. T.**, Lee, J., Suk, J. W.*, Lee, W.* "Interface Engineering to Improve Electrochemical Performance of Intermediate-Temperature Solid Oxide Fuel Cells." *Applied Surface Science* 639 (2023), 158188.
15. Nguyen, D. D.†, **Megra, Y. T.†**, Lim, S., Lim, T., Suk, J. W. "Tunable Interlayer Interactions in Reduced Graphene Oxide Paper." *ACS Applied Materials & Interfaces* 15 (5) (2023), 7610–7626. [Equal contribution.](#)

2022

16. Kim, S. J.†, Choi, M.†, **Megra, Y. T.**, Yoo, H. S., Lee, J., Suk, J. W., Lee, W. "Precursor-Driven Facile Densification of Gd_{0.1}Ce_{0.9}O_{2-δ} Interlayer for High-Performance Solid Oxide Fuel Cells." *International Journal of Energy Research* 46 (12) (2022), 17511–17520.

2021

17. **Megra, Y. T.**, Lim, S., Lim, T., Na, S. R., Suk, J. W.* "Enhancement of the Adhesion Energy Between Monolayer Graphene and SiO₂ by Thermal Annealing." *Applied Surface Science* 570 (2021), 151243.

18. Seok, H.†, **Megra, Y. T.†**, Kanade, C. K.†, Cho, J., Kanade, V. K., Kim, M., Lee, I., Yoo, P. J., Kim, H.-U.*, Suk, J. W.*, Kim, T.* “Low-Temperature Synthesis of Wafer-Scale MoS₂-WS₂ Vertical Heterostructures by Single-Step Penetrative Plasma Sulfurization.” ACS Nano 15 (1) (2021), 707–718. **Equal contribution.**

2019

19. **Megra, Y. T.**, Suk, J. W.* “Adhesion Properties of 2D Materials.” Journal of Physics D: Applied Physics 52 (2019), 364002.
20. Kang, H., Zhao, C., Huang, J., Ho, D. H., **Megra, Y. T.**, Suk, J. W., Sun, J., Wang, Z. L., Sun, Q.*, Cho, J. H.* “Fingerprint-Inspired Conducting Hierarchical Wrinkles for Energy-Harvesting E-Skin.” Advanced Functional Materials 29 (2019), 1903580.

2018

21. Betelie, A. A.*, **Megra, Y. T.**, Redda, D. T., Sinclair, A. “Experimental Investigation of Fracture Toughness for Treated Sisal Epoxy Composite.” AIMS Materials Science 5 (1) (2018), 93–104.

CONFERENCE PAPERS

22. Seo, B. H., **Megra, Y. T.**, Suk, J. W. “Measurements of the Adhesion Energy of CVD-Grown Monolayer Graphene on Dielectric Substrates.” Composites Research 36 (5) (2023), 377–382.
23. Yoo, H. S., Kim, S. J., **Megra, Y. T.**, Lee, J., Suk, J. W., Lee, W. “Improvement of Solid Oxide Fuel Cells Performance by Enlarging Contact Area at the Interface Using Electrostatic Spray Deposition.” (2023).
24. **Megra, Y. T.**, Suk, J. W. “Delamination of Large-Area CVD-Graphene from SiO₂.” Korean Society of Mechanical Engineers Conference, 2020.
25. **Megra, Y. T.**, Nguyen, D. D., Suk, J. W. “Double Cantilever Beam (DCB) Tests of Graphene Papers.” Korean Society of Mechanical Engineers Conference, 2019.
26. **Megra, Y. T.**, Suk, J. W. “Finite Element Analysis of Adhesion Properties of Graphene on SiO₂.” Korean Society of Mechanical Engineers Conference, 2018.

AWARDS & SCHOLARSHIPS

- Excellent Poster Presentation Award, 2023 Conference of the Korean Society for Composite Materials, for research on graphene adhesion according to underlying substrate characteristics.
- Excellent Presentation Award, 2022 Fall Conference of the Korean Society for Composite Materials, for “Measurement of the Adhesion Energy of Monolayer Graphene Using Mode-I Fracture Test.”
- Excellent Oral Presenter Award, Mechanical Engineering Graduate School Academic Festival, Sungkyunkwan University, January 2022, for “Controlling Interlayer Cohesion Energy in Graphene Papers through Chemical Modification.”
- Postgraduate Scholarship, Ph.D. program, Sungkyunkwan University, 2017–2019.
- Postgraduate Scholarship, M.Sc. program, Addis Ababa University, 2015–2017.

SELECTED CONFERENCE PRESENTATIONS

Oral

1. Controlling Interlayer Cohesion Energy in Graphene Papers through Chemical Modification. 66th Winter Annual Conference of the Korean Vacuum Society, Gangwon-do, Republic of Korea, February 14–16, 2024.
2. Measurements of the Adhesion Energy of Monolayer Graphene According to the Surface Characteristics of SiO₂/Si Substrates. 66th Winter Annual Conference of the Korean Vacuum Society, Gangwon-do, Republic of Korea, February 14–16, 2024.
3. Effect of Reduction Degree on the Interface Delamination of Graphene Oxide Papers. Mechanical Engineering Graduate School Academic Festival, Sungkyunkwan University, January 14, 2022. Excellent Presenter Award.
4. Interfacial Adhesion Between Large-Area Monolayer Graphene and SiO₂/Si. 8th International Conference on Microelectronics and Plasma Technology and 9th International Symposium on Functional Materials, Republic of Korea, January 17–20, 2021.

Poster

1. Spectrometer-on-a-Barristor for High Reconstruction Fidelity with Multidimensionally-Encoded Spectral Responsivity. 22nd International Symposium on the Physics of Semiconductors and Applications, Jeju, Republic of Korea, June 28–July 2, 2026.
2. Topological Dielectrics for Two-Dimensional Nanoelectronics. 22nd International Symposium on the Physics of Semiconductors and Applications, Jeju, Republic of Korea, June 28–July 2, 2026.
3. Functionalized Topological Insulating van der Waals Dielectrics. 32nd Korean Conference on Semiconductors, Gangwon, Republic of Korea, February 14, 2024.
4. Universal Logic Gates Achieved by a Single-Channel Graphene Device. 32nd Korean Conference on Semiconductors, Gangwon, Republic of Korea, February 14, 2024.
5. Tellurium Based p-Type Material Growth for Electronic Device. 32nd Korean Conference on Semiconductors, Republic of Korea, February 13, 2024.
6. Mechanical Separation of Free-Standing Graphene Oxide Papers. 2023 Conference of the Korean Society for Composite Materials, Geoje-si, Republic of Korea, November 8–10, 2023.
7. Measurement of the Adhesion Energy of Graphene According to the Surface Characteristics of the Underlying Substrates. 2023 Conference of the Korean Society for Composite Materials, Geoje-si, Republic of Korea, November 8–10, 2023. Excellent Poster Presenter Award.

8. Conductive and Stretchable Fibers for Wearable Strain Sensors Using Carbon Nanofiber-Based 3D Structure. NANO KOREA 2023, KINTEX, Republic of Korea, July 5–7, 2023.
9. Interface Resistance Analysis in Solid Oxide Fuel Cells. 243rd ECS Meeting with SOFC-XVIII, Boston, MA, USA, May 28–June 2, 2023.
10. Analysis of Adhesion Characteristics Between Monolayer Graphene and a Substrate. 2023 Conference of the Korean Society for Composite Materials, Jeju-do, Republic of Korea, April 19–21, 2023.
11. Vertical Graphene Grown on Carbon Nanofibers for Energy Storage Device Electrodes. 2022 Conference of the Korean Society of Mechanical Engineers, Jeju-do, Republic of Korea, November 9–12, 2022.
12. Measurement of the Adhesion Energy of Monolayer Graphene Using Mode-I Fracture Test. 2022 Fall Conference of the Korean Society for Composite Materials, November 4, 2022.
13. Modification of the Adhesion Energy of Monolayer Graphene on SiO₂. 2022 Conference of the Korean Society of Mechanical Engineers, Busan, Republic of Korea, May 18–20, 2022.
14. Bilayer Graphene Synthesis on Copper Using Conventional Thermal CVD. 2022 Conference of the Korean Society for Composite Materials, Jeju-do, Republic of Korea, May 18–20, 2022.
15. Delamination of Large-Area CVD-Graphene from SiO₂. Korean Society of Mechanical Engineers Fall Conference, Republic of Korea, December 2020.
16. Interfacial Delamination of Graphene Paper. 2019 Conference of the Korean Society of Mechanical Engineers, Jeju-do, Republic of Korea, November 13–15, 2019.
17. Mechanical Delamination of Monolayer Graphene from Dielectrics. 17th International Nanotech Symposium & Exhibition (NANO KOREA), KINTEX, Ilsan, Republic of Korea, July 3–5, 2019.
18. Double Cantilever Beam (DCB) Tests of Graphene Papers. Korean Society of Mechanical Engineers Fall Conference, Republic of Korea, 2019.
19. Finite Element Analysis of Adhesion Properties of Graphene on SiO₂. Korean Society of Mechanical Engineers Spring Conference, Republic of Korea, 2018.

TECHNICAL EXPERTISE

- **2D Materials & Synthesis:** Graphene; MoS₂; WS₂; WSe₂; MXenes; 2D heterostructures; CVD; PECVD; ALD; surface mono-crystallization of Cu; wet transfer; defect engineering; surface/interface functionalization.
- **Nanofabrication:** Electron-beam lithography; electron-beam evaporation; reactive-ion etching; mask alignment; spin coating; dip coating; nanogap fabrication; device fabrication; semiconductor-device integration.
- **Characterization:** Raman; AFM; SEM/FESEM; TEM; STM; XPS; XRD; FTIR; EDS; EBSD; optical microscopy; surface profiling.
- **Molecular Electronics:** STM break-junction; EC-STM; cyclic voltammetry; single-molecule charge transport; molecular conductance; quantum interference; molecular/2D interfaces; I–V/C–V measurements.
- **Interface Mechanics:** Mode-I fracture mechanics; DCB testing; adhesion/cohesion energy; interlayer interactions; delamination; self-healing interfaces; bulge testing; traction–separation relations; ANSYS; Abaqus.
- **AI & Computational Research:** AI-assisted materials research; data-driven materials analysis; computational materials approaches; finite-element modeling; integration of functional materials with AI-enabled sensing and semiconductor devices.

PROFESSIONAL & RESEARCH CONTRIBUTIONS

- Interdisciplinary research across materials science, chemistry, mechanical engineering, semiconductor engineering, nanotechnology, molecular electronics, quantum transport, computational materials science, and energy materials.
- Peer-review activities for journals in materials science, nanotechnology, 2D materials, energy materials, and related fields.
- Undergraduate research supervision and mentoring; graduate student research guidance.
- Undergraduate laboratory teaching, assessment, and laboratory-manual development.
- Research infrastructure and laboratory-planning contributions at XJTLU.

RESEARCH VISION

Molecularly Engineered 2D Materials for Quantum and Low-Power Electronics

Central question: How can molecular and interfacial engineering be used to control charge transport and emergent electronic phenomena in two-dimensional materials?

- Molecular and interface engineering
- 2D-material synthesis and nanofabrication
- Quantum and molecular transport
- Semiconductor device integration
- Advanced characterization
- Data-driven and AI-assisted materials research

Core concept: Treat the interface as an active design element rather than a passive boundary between materials.

REFERENCES

Prof. Li Yang — Associate Dean for Research; Advanced Materials Research Center, XJTLU
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Professor John Moraros — Dean, School of Science, XJTLU
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Prof. Lifeng Ding — Senior Associate Professor, Academy of Life and Natural Sciences, XJTLU
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Prof. Ji Won Suk — School of Mechanical Engineering; SKKU Advanced Institute of Nanotechnology (SAINT), SKKU
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